Experiment No. 8: Creating a Dynamic Dashboard

Data Visualization and Presentation Lab • Course Code: 25UMEP304B • Roll No. 60

Domain and Dashboard

Domain: EV Charging Analytics
Dashboard name: EV Charging Analytics Dashboard
Problem statement: Analyze charging energy, cost, session status and station-level activity to support operational monitoring and resource planning.
Target user: EV charging station operator or network manager.

Three meaningful findings

1. Energy by station: Andheri GreenCharge has the highest total energy in the original 69-session dataset (400.4 kWh). The operator can review its utilization and compare demand with other stations when planning capacity.

2. Cost by vehicle type: Electric SUV contributes the highest total charging cost (9,984 INR) in the original dataset. This can help managers compare charging demand and revenue contribution by vehicle category.

3. Session outcomes: The dataset records 56 completed sessions and 13 cancelled sessions out of 69 sessions. Tracking cancellations alongside completed sessions can help identify stations or periods that may need service-quality investigation.

Dashboard interactivity and refresh test

The report includes KPI cards for total energy, total charging cost and session count; charts for energy by station, cost by vehicle type and sessions by status; and City and Vehicle Type slicers. Select one slicer value at a time and capture a screenshot showing how the cards and charts respond. For the refresh demonstration, the final Excel workbook contains five additional records (EV070–EV074), increasing the dataset from 69 to 74 rows. In Power BI Desktop, update the source to this final workbook if required, select Home → Refresh, and capture the before/after views. The screenshots must be captured from the actual Power BI application.

Brief conclusion

The dashboard consolidates charging activity into a small set of KPIs and breakdowns that help a station operator compare energy demand, charging cost and session outcomes. City and Vehicle Type slicers support focused exploration. The findings are descriptive of the supplied sample and should be rechecked after refreshing the model with the five added records.

Refresh records added to the final dataset

Session_ID	Date	Station	Vehicle	Energy (kWh)	Cost (INR)
EV070	2026-09-24	Pune Central EV Hub	Electric Car	31.5	504
EV071	2026-09-25	Andheri GreenCharge	Electric SUV	42.0	672
EV072	2026-09-26	Nashik Road ChargeHub	Electric 2-Wheeler	6.8	109
EV073	2026-09-27	Thane UrbanCharge	Electric Car	27.4	438
EV074	2026-09-28	Vashi EV Plaza	Electric SUV	36.2	579

Submission reminder: Submit the final Excel dataset, actual Power BI .pbix file, actual dashboard screenshot, slicer/filter interaction screenshots, before/after refresh screenshots, and this findings/conclusion PDF. This PDF does not replace the PBIX or application screenshots.